Computer Science & IT

Theory of Computation



CFL

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



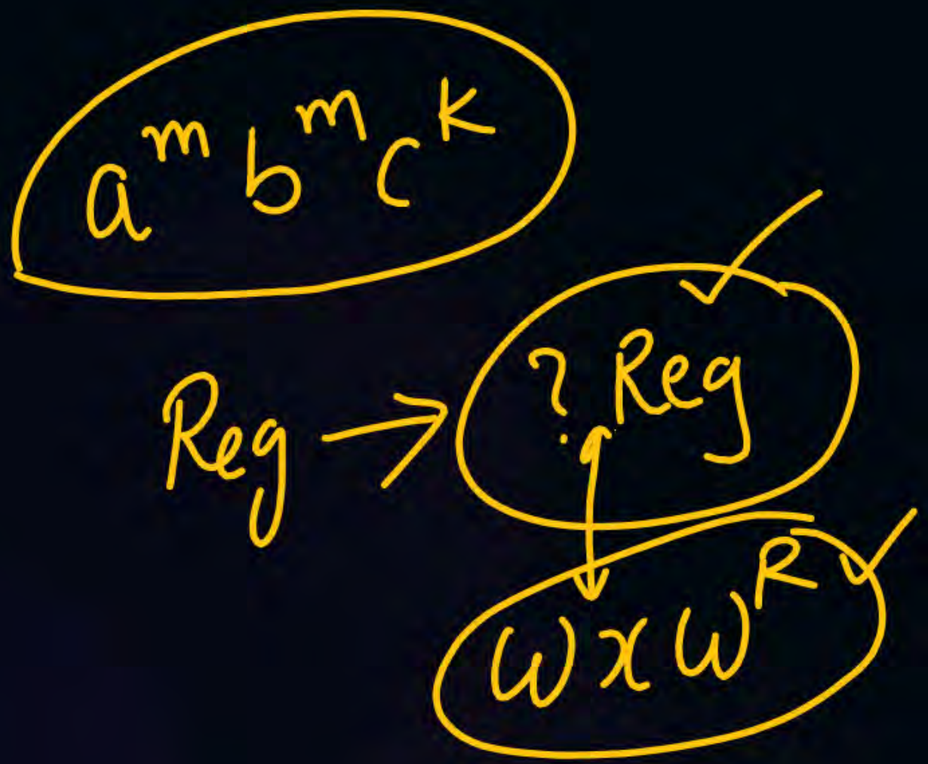
- Topic
- Topic
- Topic

m, m

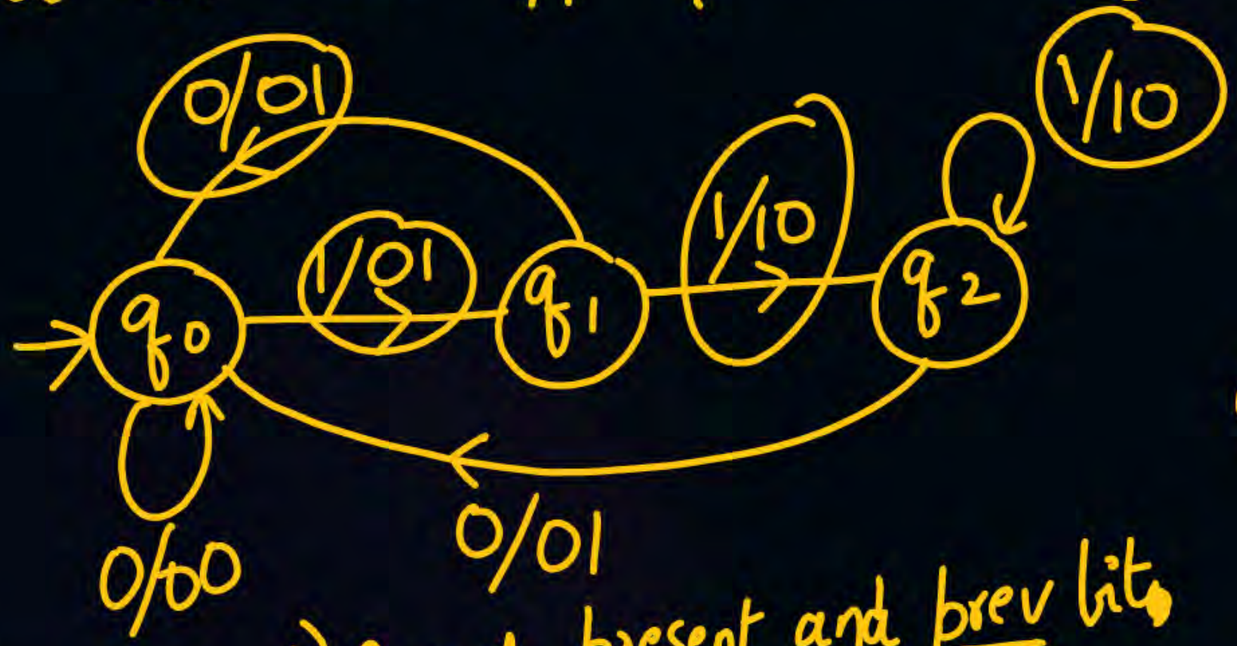
Topics to be Covered

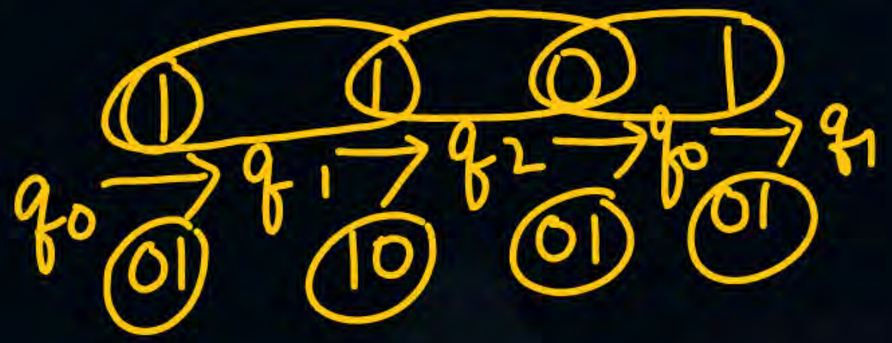


- Topic
- Topic
- Topic


$a^m b^m c^k$
Reg →
? Reg
$w x w^R$

what is the o/p produced by following m/c





a) $11 \rightarrow 01$

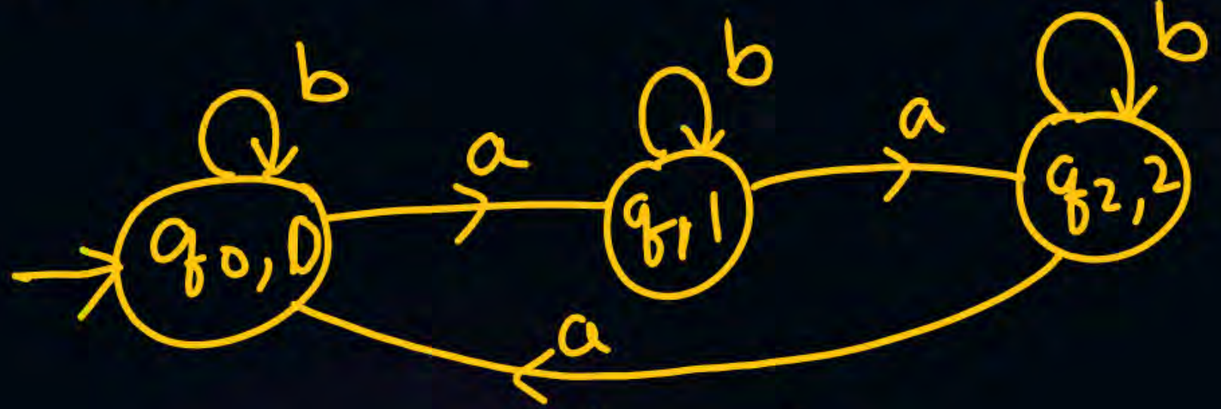
b) Sum of present and prev bits

c) $10 \rightarrow 00$

~~d)~~ none

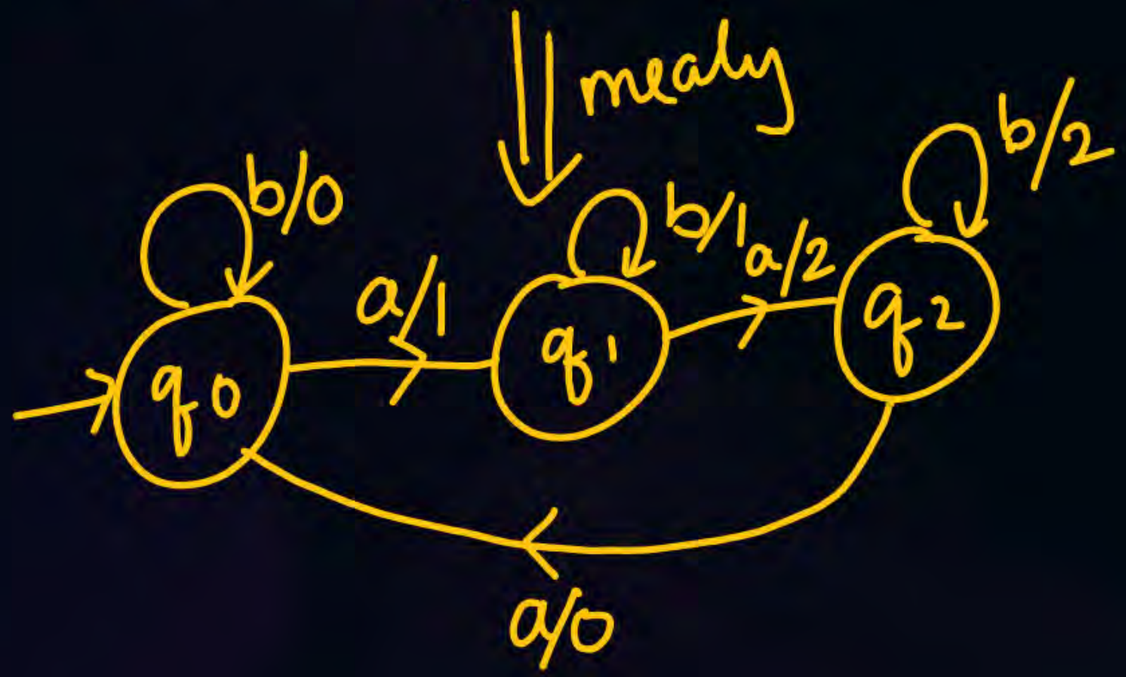
mealy

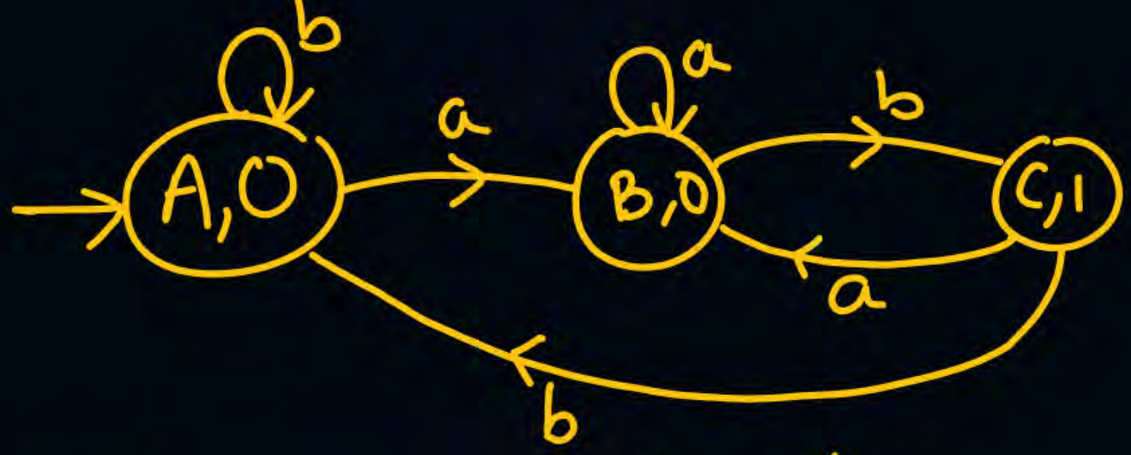
Construct a mode m/c Couting no of a's mod 3. $\Sigma = \{a, b\}$



moore → State + O/P

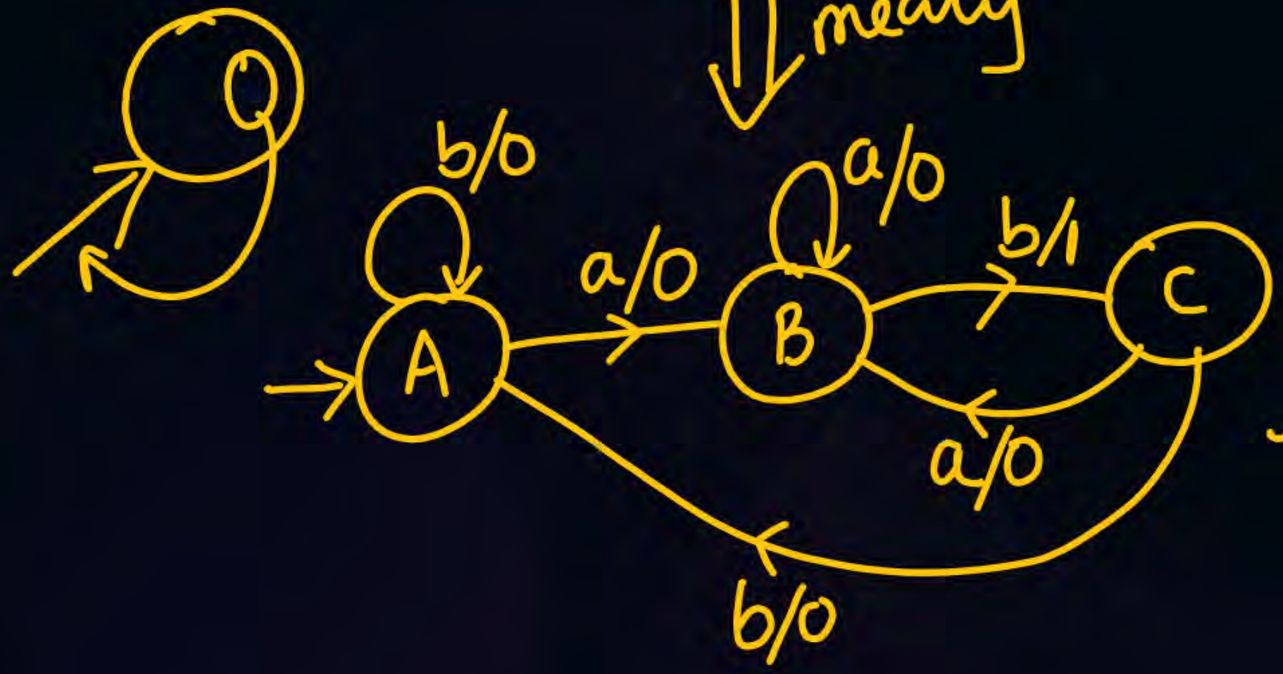
mealy → Trans + O/P



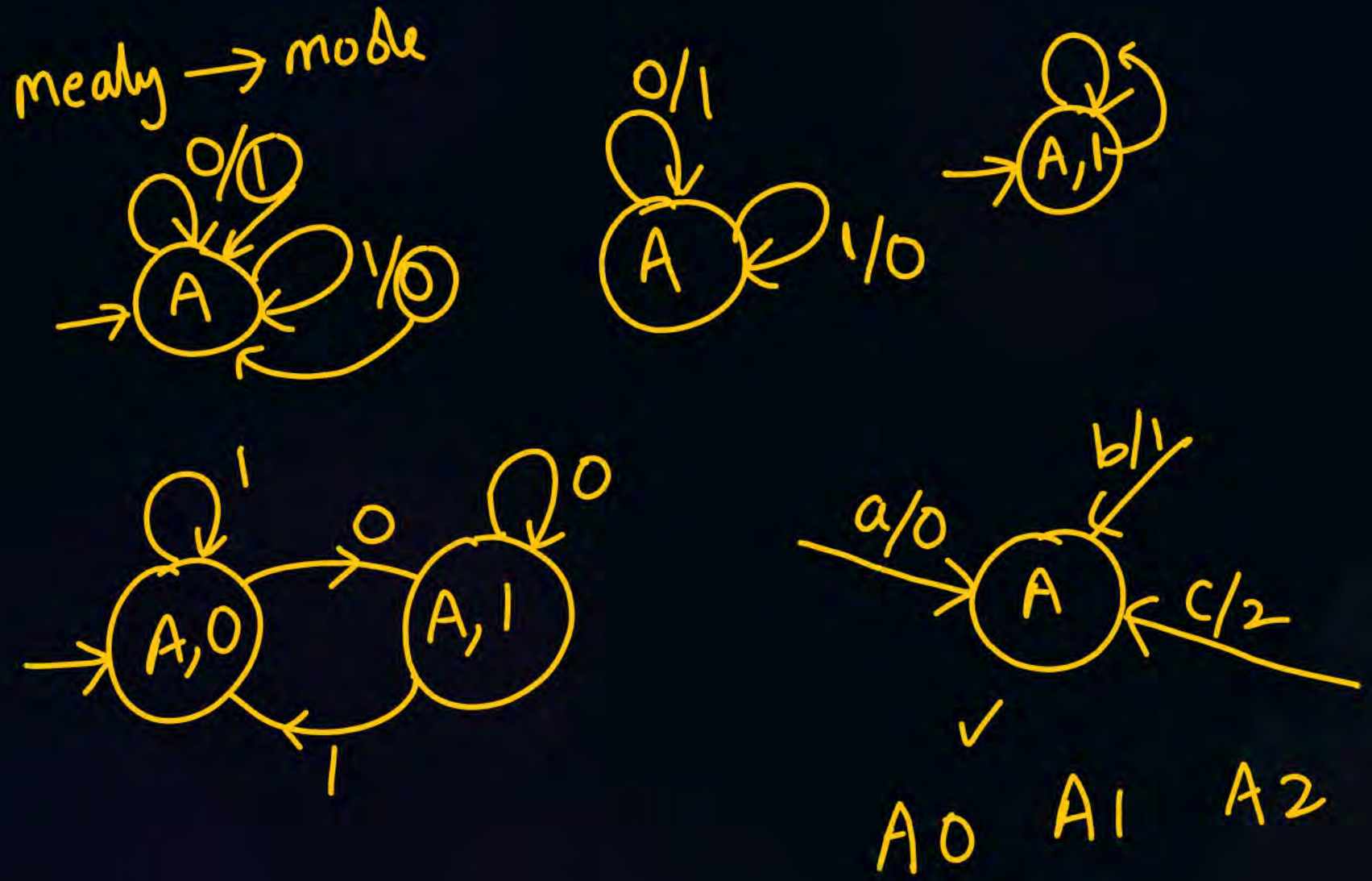


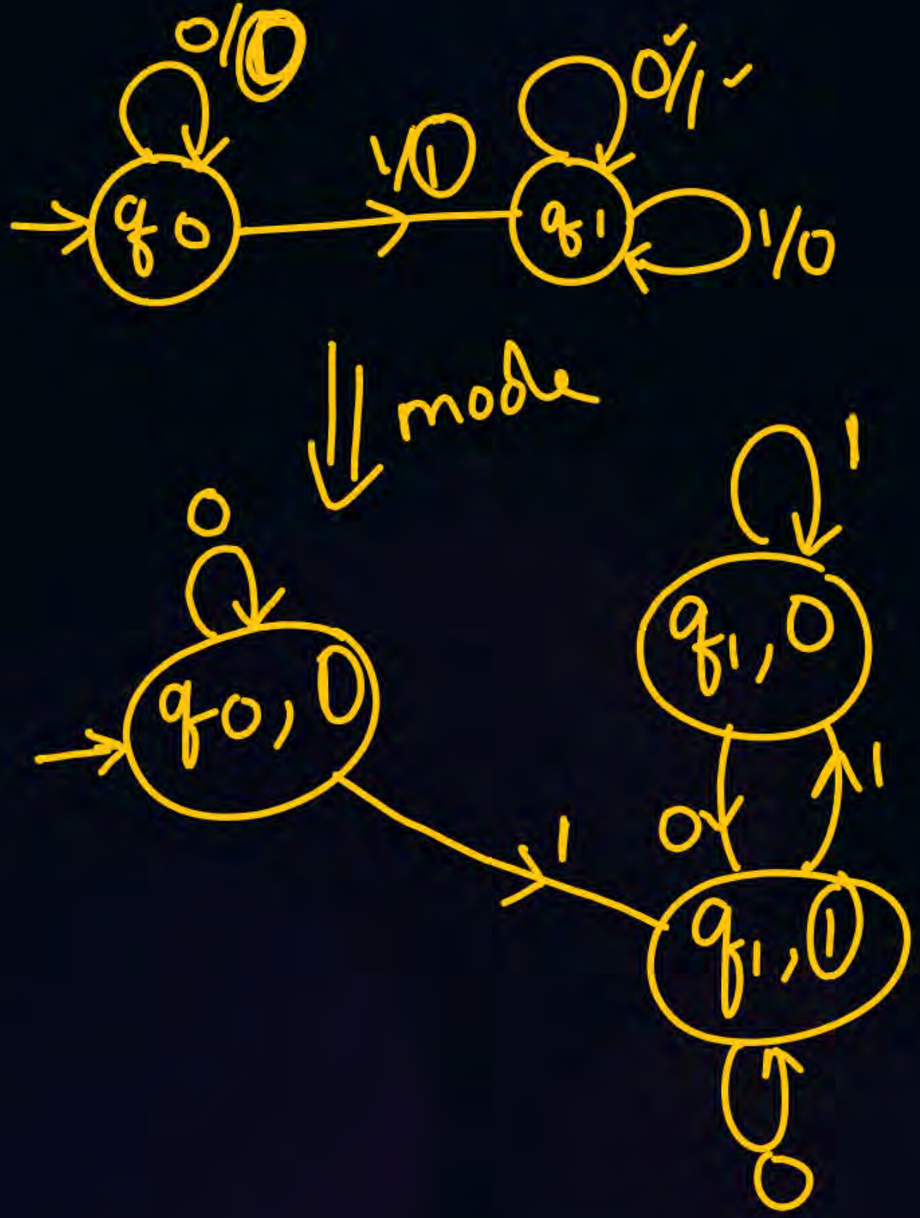


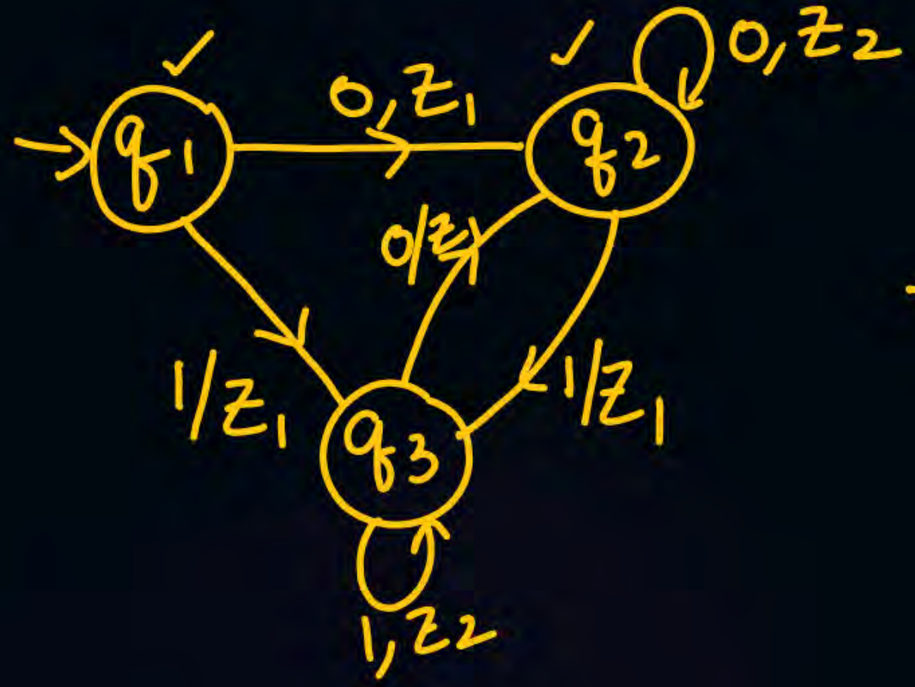
mealy

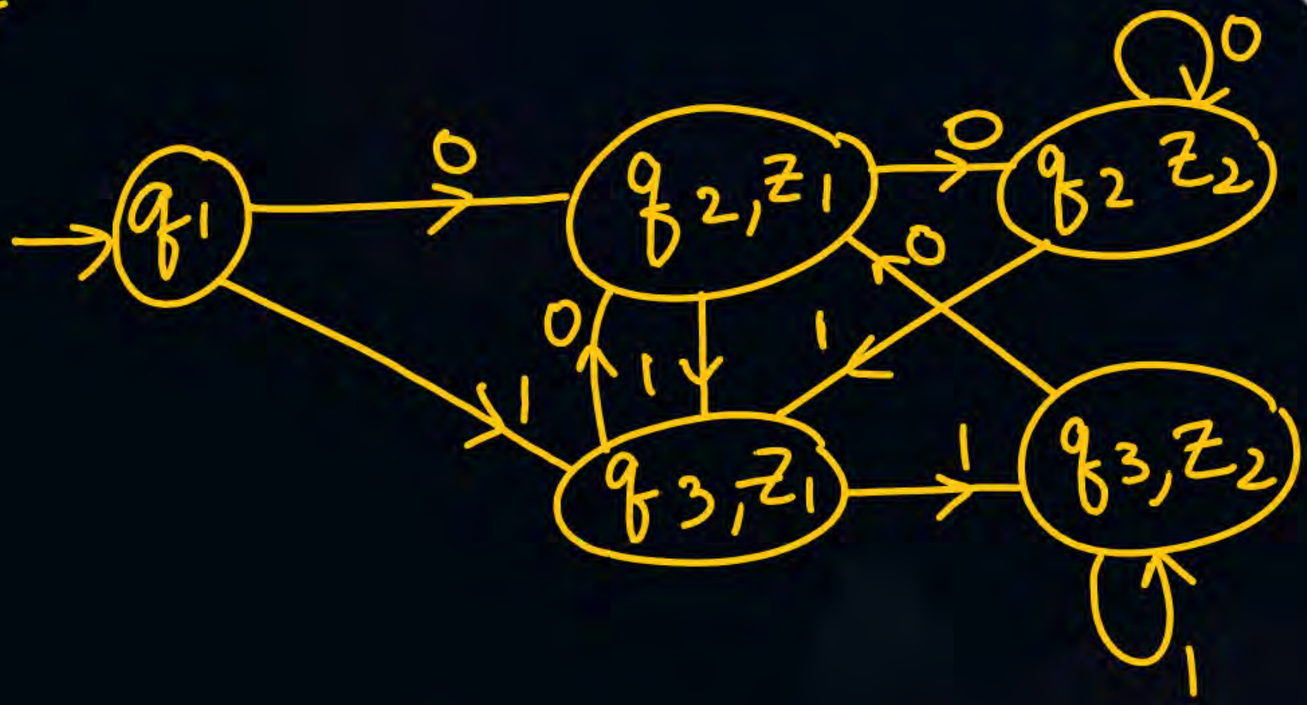


incoming edge will get the state o/p


Mealy → mode
0/1
1/0
A
0/1
1/0
A
A,1
1
0
0
A,0
A,1
1
b/1
a/0
A
c/2
A0 A1 A2


0/0
1/1
0/1
1/0
q0
q1
mode
0
1
q0,0
q1,0
q1,1
0
1
0
1


q1
q2
q3
0, Z1
0, Z2
0/Z1
1/Z1
1/Z1
1, Z2


q1
q2, Z1
q2 Z2
q3, Z1
q3, Z2
0
1

Reg - 90%

$(Q, \Sigma, \delta, q_0, \Delta, \lambda)$

- Q - Finite set of states
- Σ - I/p alphabet
- δ - Transition function ($\delta : Q \times \Sigma \to Q$)
- q_0 - initial state
- Δ - O/p alphabet
- λ - O/p function

MOORE
one output

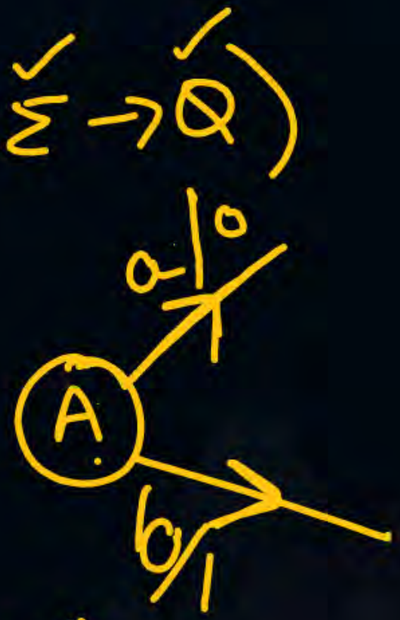
WxW^R

Moore m/c

$\lambda : Q \to \Delta$

Mealy m/c

$\lambda : Q \times \Sigma \to \Delta$



A, a →

$\Sigma = \{a, b\}$

A, b →

Gate: Reg

Lang → RE

RE → Lang

Lang → DFA

Lang — min FA ✓

DFA → Lang

NFA → Lang

Reg & not ✓

min of DFA

Arden's Lemma

Context free languages :-

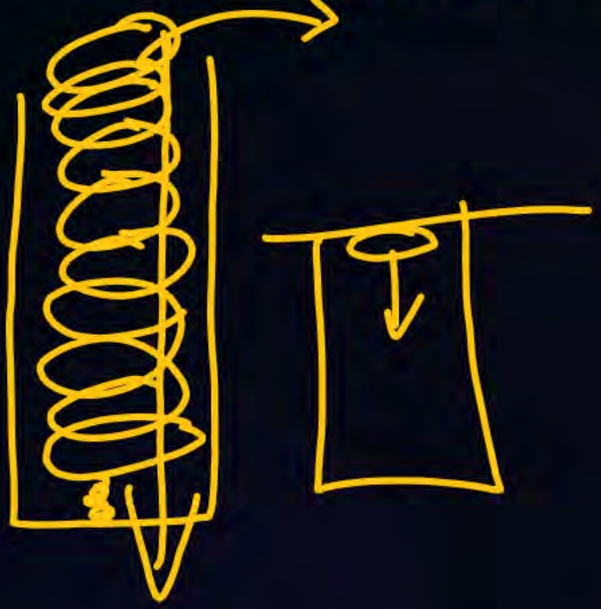
RG
CFG
CSG
URG

CFG

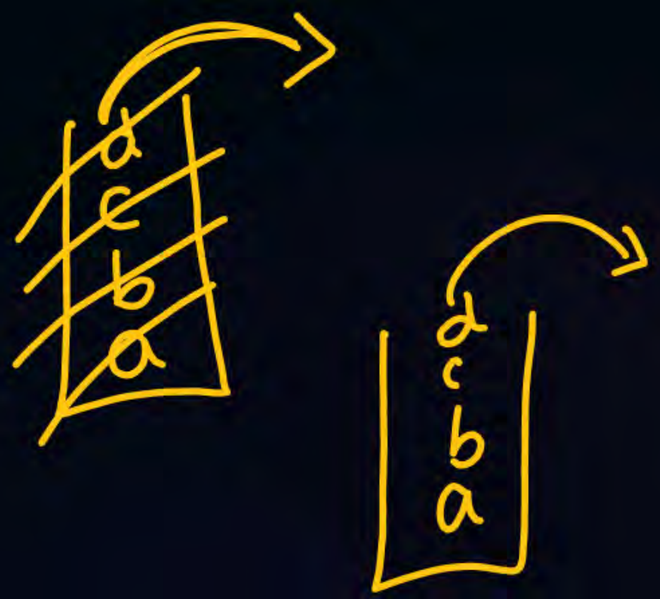
Context free grammars

PDA

Push down automata



PDA = FA + stack ✓


d
c
b
a

d
c
b
a

Types of grammar :- (chomsky hierarchy)

Type 0 grammars - (unrestricted grammars)

These grammars include all formal grammars.
In unrestricted grammars all productions are of form
$\alpha \to \beta$ where α and β may have any number
of terminals and non terminals with atleast one non terminal
on the LHS.
The language

$aAb \to ab$

$Aa \to aAb$

$A \to \epsilon$

$AB \to \epsilon$

Ex of type 0 grammar:- (Standard text book example)

$L(G) = \{a^{2^i} / i \geq 1\}$

$S \rightarrow ACaB$

$Ca \rightarrow aaC$

$CB \rightarrow DB$

$CB \rightarrow E$

$aD \rightarrow Da$

$AD \rightarrow AC$

$aE \Rightarrow Ea$

$AE \rightarrow \epsilon$

$S \Rightarrow ACaB \Rightarrow AaaCB \Rightarrow AaaE \Rightarrow AaEa$

$\Rightarrow AEaa$

$\Rightarrow aa$

Type 1 grammar:-

These grammars define CSL.

All productions are of form $\alpha \to \beta$ where $|\alpha| \leq |\beta|$, α, β may have any number of terminals and non terminals. α should have atleast one variable

It cannot have the production $A \to \epsilon$.

These are called context sensitive grammars.

Because $\alpha A \beta \to \alpha \gamma \beta$, A is replaced by γ only when in the context A appears b/w α and β

$|A| \to |\epsilon|$ X

$1 \leq x$ 0

$|aA| \to |ab|$

2 2

$|aAb| \to |aab|$

3 ≤ x 2

$\Sigma = \{\ \}$ ↓ Terminals

non terminal & variable

Ex of Type 1 grammar (Standard text book)

$L(G) = a^n b^n / n \geq 1$

$S \rightarrow \underline{SBC} / ac$

$B \rightarrow a$

$CB \rightarrow BC$

$Ba \Rightarrow aa$

$C \rightarrow b$

$S \Rightarrow \underline{S}BC \Rightarrow SBCBC \Rightarrow aCBCBC$

$\Rightarrow aBCCBC \Rightarrow aaCCBC \Rightarrow aaCBCC$

$\Rightarrow aaBCC \Rightarrow aaaCCC \Rightarrow \underline{aaabbb}$

Type 2 grammars (Context free grammars)

These grammars define Context free languages

They are defined by rules of form $\alpha \to \beta$ with $|\alpha| \leq |\beta|$

and $|\alpha| = 1$ and (α is non terminal). $\alpha \to \beta$, we can replace

α by β in any context, so it is context free.

$|A| \to |\in|$

1 X 0

CFG ✓

CSG ✓

$A \to \in$

S → aS

A → bA

B → cB

non terminals

$\Sigma = \{a, b, c\}$

terminals

$\alpha \to \beta$ $|\alpha| \leq |\beta|$

$|Aa| \to |cA|$

$|Aba| \to |Aa|$ ✓ ✓

3 ✗ 2

$|Aa| \to |Aba|$

$2 \leq$ ✓ 3

$\Sigma = \{a, b\}$ ✓

$S \rightarrow aA$

$A \rightarrow bB$

$B \rightarrow aB$ /

$T_\Sigma = \{a, b\}$

non terminal or variable $= \{S, A, B\}$

$L = \{a^n b^n / n \geq 1\}$ ✓

$S \rightarrow aSb / ab$ ✓

Type 3 grammar (Regular grammar)

These grammars generate regular languages

- Right linear
  $A \to \alpha X / \beta$
- Left linear
  $A \to X\alpha / \beta$

$\alpha, \beta \in T^*$
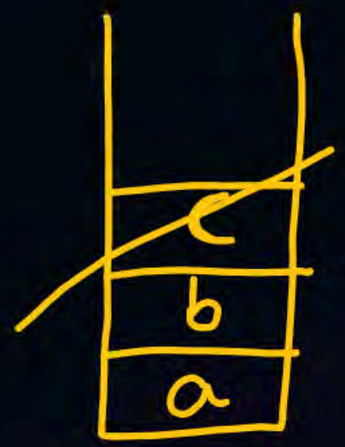
A, X are variables

$S \to \epsilon$ is allowed

$A \to \beta$ $\quad \beta \in T^*$

Type	grammar	Language	Corresponding m/c
Type 0	unrestricted grammars	Recursively enumerable	Turing machine
Type 1	Context Sensitive grammar	Context Sensitive language	Linear Bounded automata
Type 2	Context free grammar	Context free languages	Pushdown automata
Type 3	Regular grammar	Regular languages	Finite automata

PDA



Top = ? c

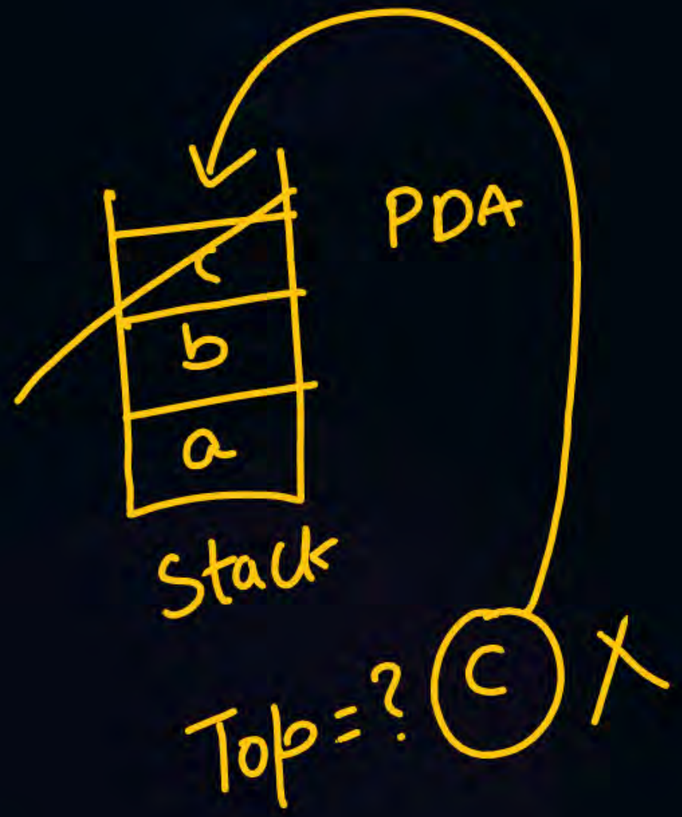
pop = c



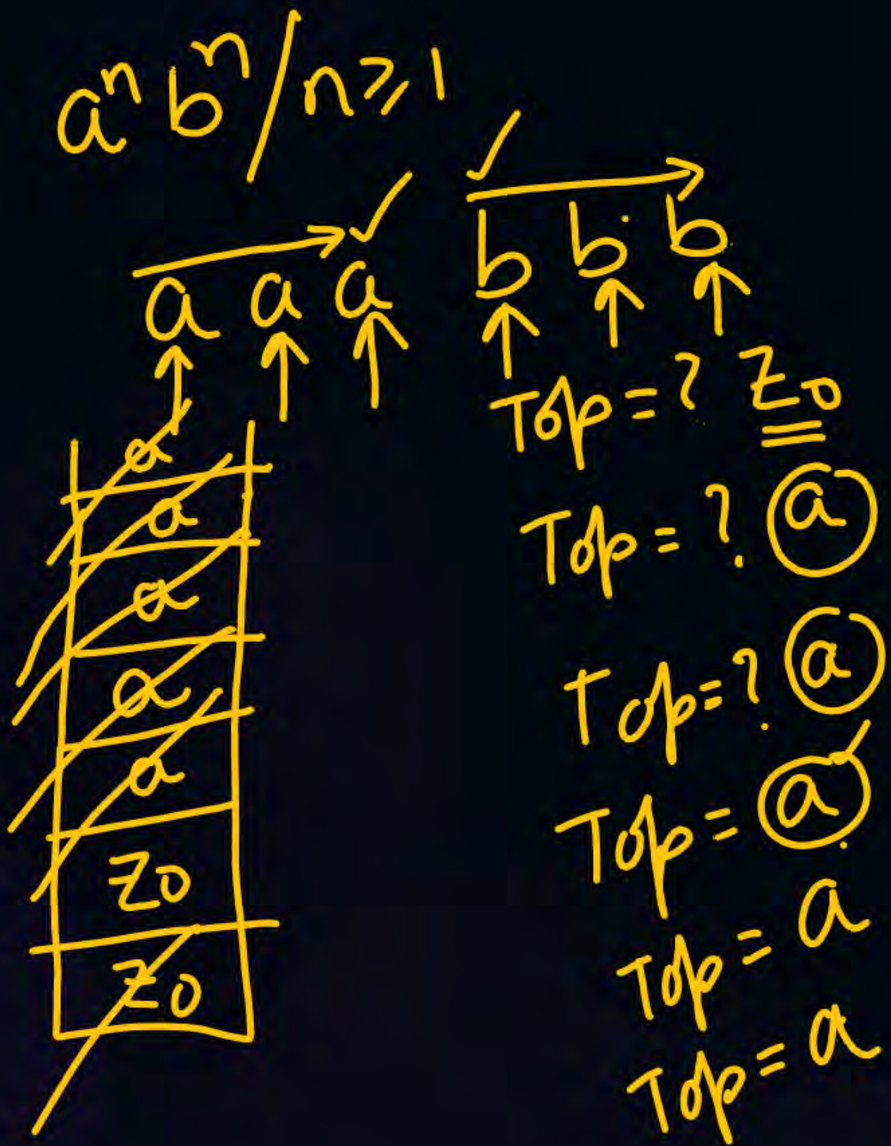
Top = ? c

pop

Top = ? b


PDA
c
b
a
Stack
Top = ?
c
X

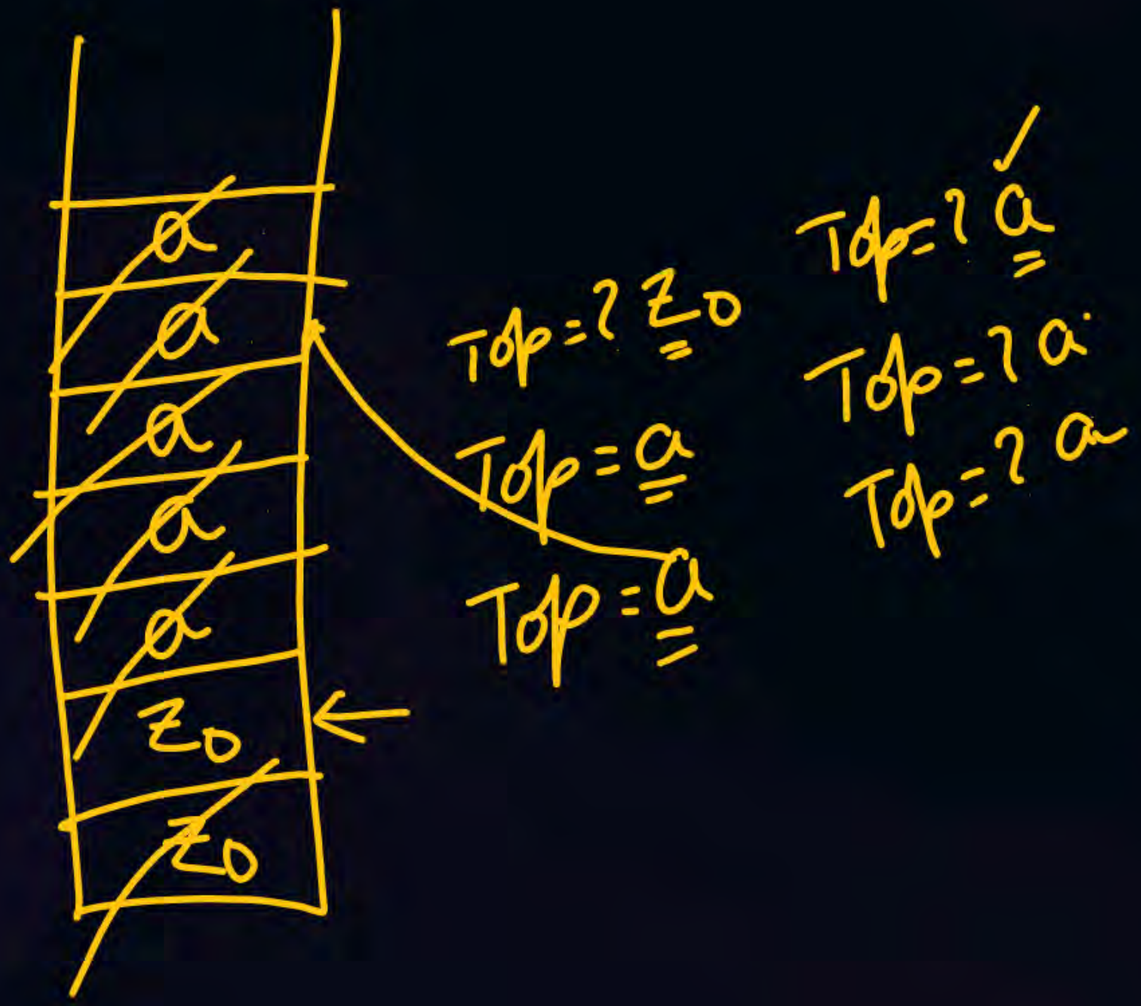
$a^n b^n / n \geq 1$



a a a b b b ∈



a a a b b b ε



a a b b ∈

PDA → pop X

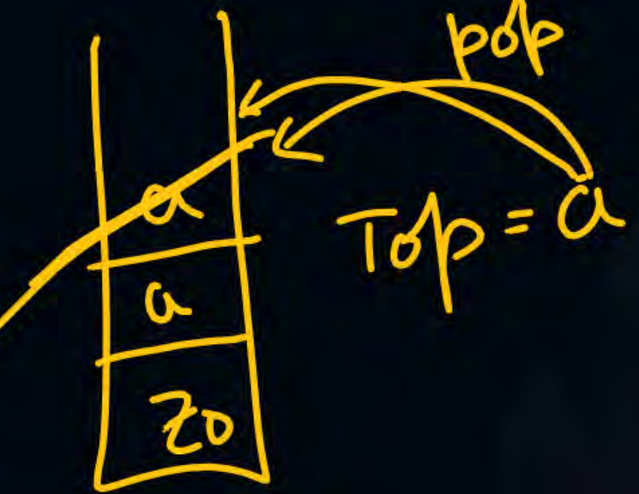


Top = a

Top = a

Top = Z0

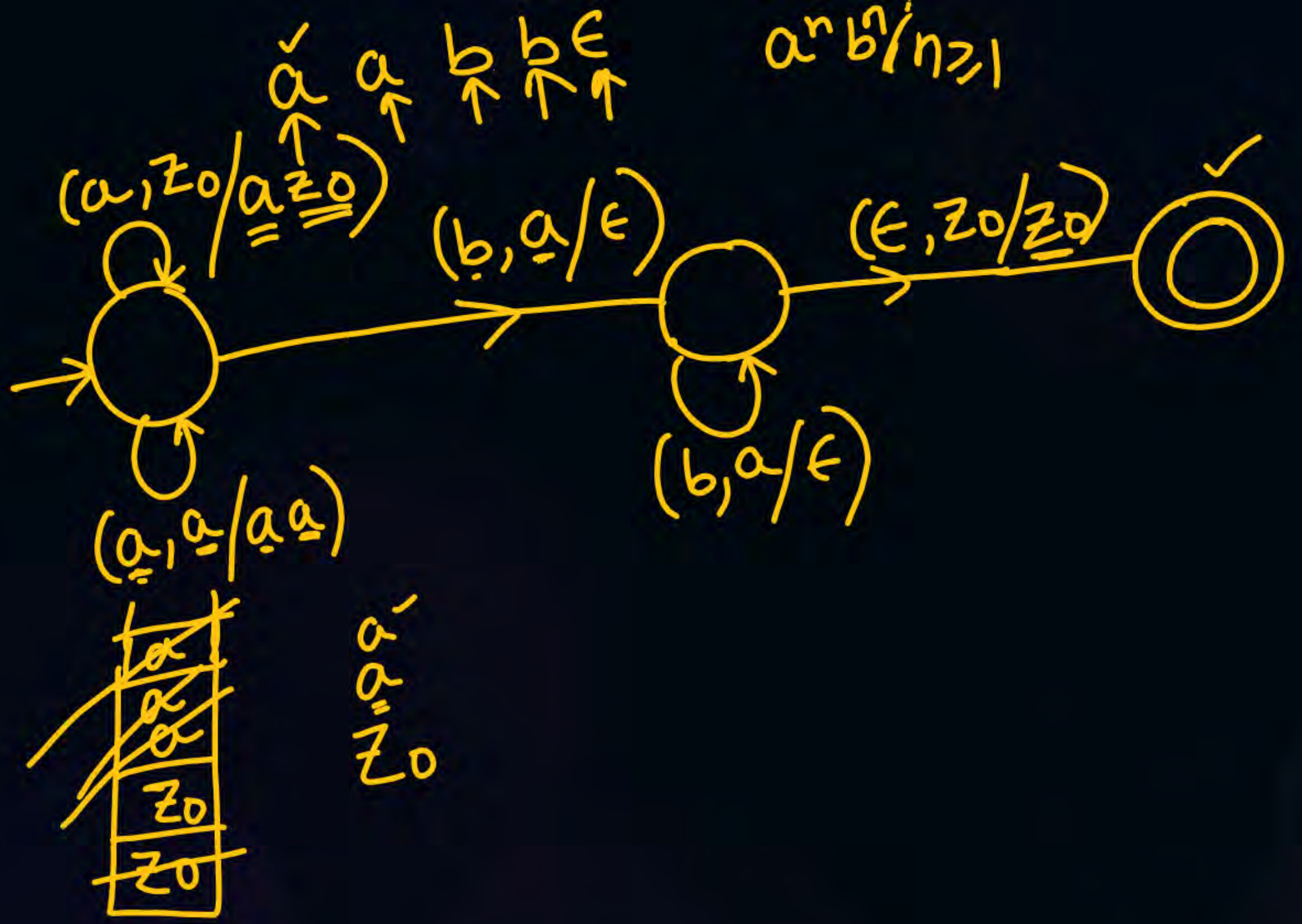
Top = α

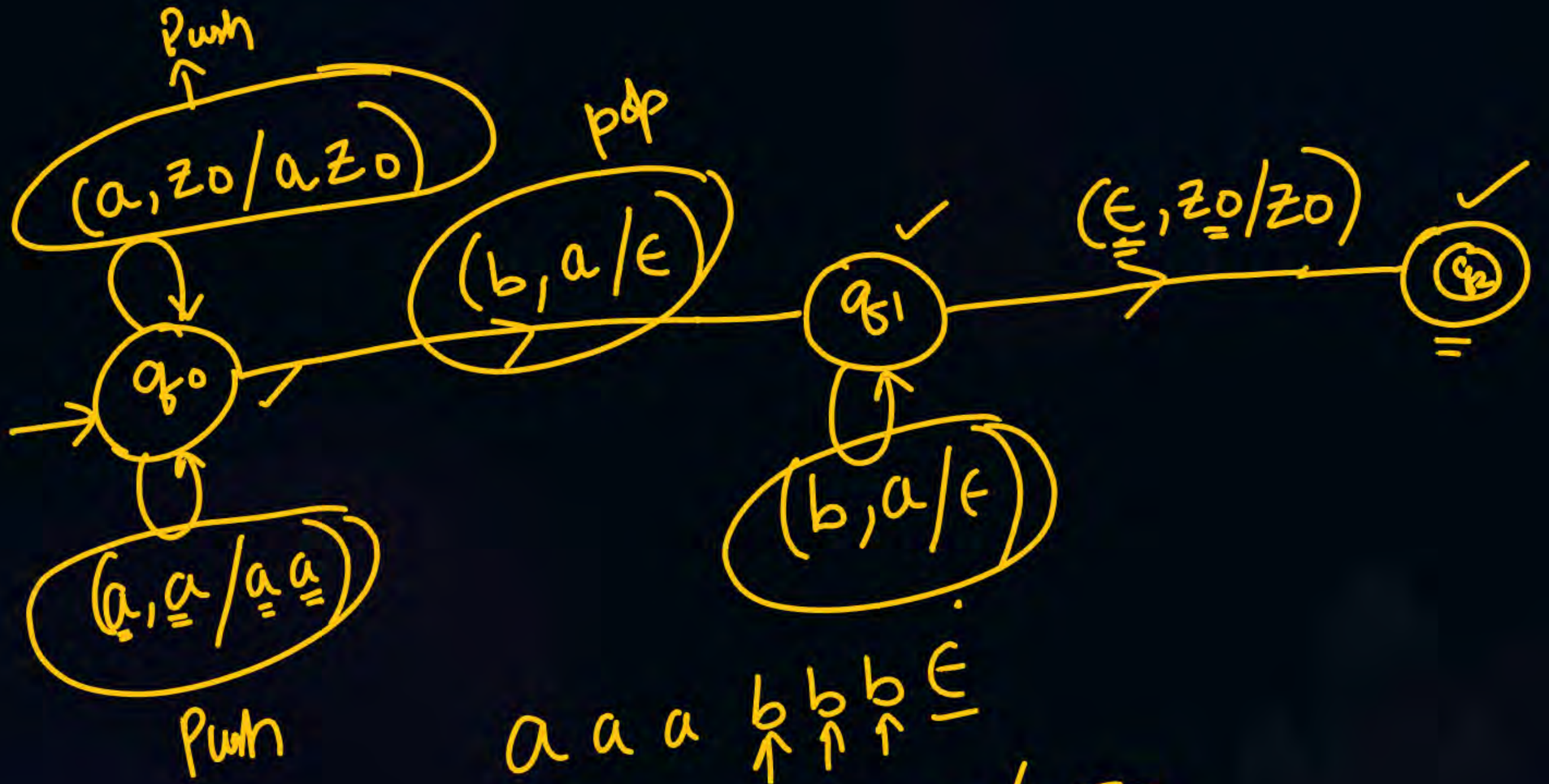


Top = a

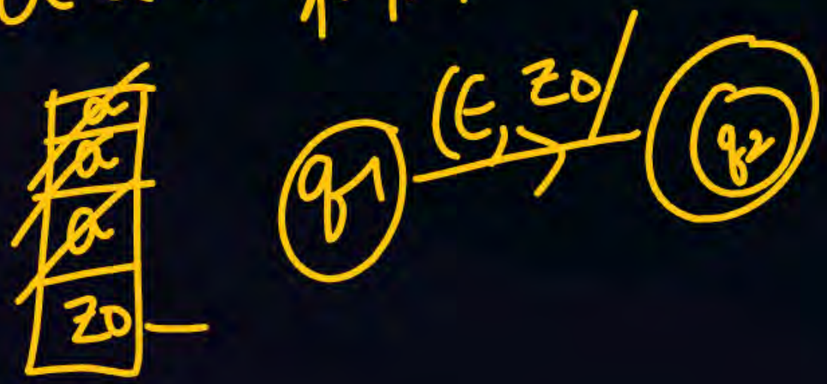
aaaabbbb



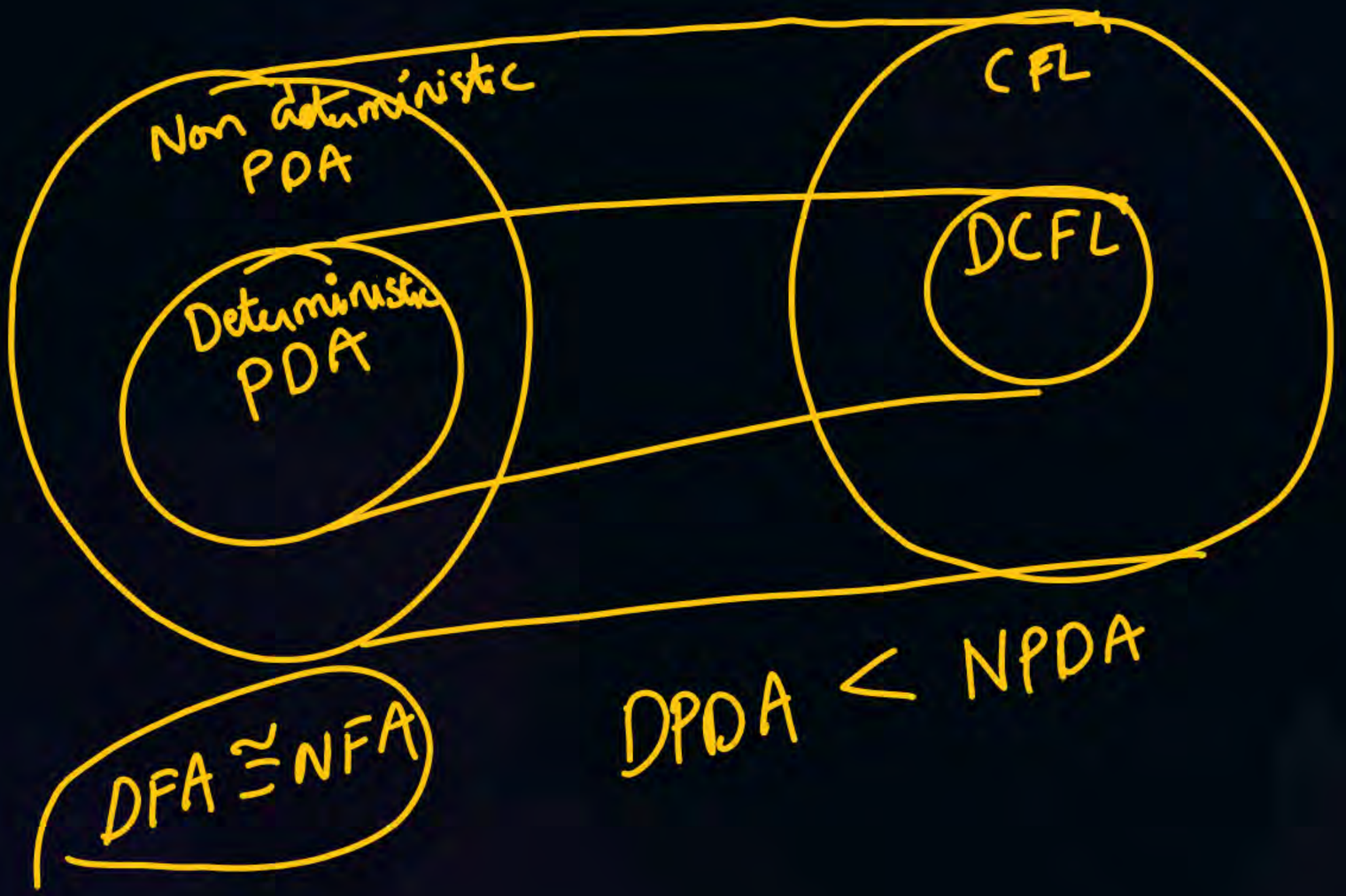

a a a b b ε
$a^n b^n / n \geq 1$
(a, Z0/a a Z0)
(b, a/ε)
(ε, Z0/Z0)
(b, a/ε)
(a, a/a a)
a a a Z0 Z0
a a Z0


Push
(a, z0/a z0)
pop
(b, a/ϵ)
q0
q1
(ϵ, z0/z0)
q2
(a, a/a a)
(b, a/ϵ)
Push

a a a b b b ϵ


a
a
a
z0
q1
(ϵ, z0/
q2


String
PDA
final State
accepted
Dead Configuration
Rejected


Non deterministic
PDA
Deterministic
PDA
CFL
DCFL
DFA ≅ NFA
DPDA < NPDA

THANK - YOU